

This work aims to develop two models to simulate the operation of a photovoltaic and predict electricity production using weather data, including solar radiation and system temperature. A prototype measurement system was developed to collect data from an photovoltaic installation. The obtained data set was prepared for developing linear and LSTM models simulating the operation of the installation. The models can be applied to estimate the value of electricity production for a specific installation in the future and the resulting plan the production and the redistribution of this energy.

Keywords: photovoltaic system modeling, LSTM, neural network, linear regression.

Introduction

This work describes the process of developing artificial intelligence models that simulate the operation of a photovoltaic panel system depending on the seasonal data (such as solar radiation and system operating temperature) and can be used for predicting electricity production. The problem is quite new, but there are some works that show existing models using neural or agent approaches, but rather taking into account technical aspects [Ag22], [DD26], and no one has tried to describe collecting data using a system designed for this purpose. Our approach presented in this paper focuses on solving the task from scratch, from constructing the data collection equipment, through data processing, to creating models using selected methods. To build such a system a special device has been constructed, which can store data directly from PV and monitor weather conditions. While there are computational models and formulas (presented in the next section) that describe PV panel parameters depending on changes in temperature and other parameters, they are difficult to use directly because the specific characteristics of the panels used are often unknown, as is the impact of their installation

Witold Kobiela, Newelska 6, Warsaw, 01-447, Poland, P.Kobiela@wit.edu.pl,  <https://orcid.org/0009-0001-55934-1234>

Research Institute Polish Academy of Sciences, Newelska 6, Warsaw, 01-447, Poland.

Basic concepts

Operation of a photovoltaic cell is based on the photovoltaic effect, a key phenomenon in solid-state physics that occurs in semiconductors (see [DLM24] or [DLM25]). The basis of photovoltaic phenomena is a semiconductor material, most commonly crystalline silicon (Si). In pure silicon, the valence band is filled, and the conduction band is empty. The energy required for an electron to jump between the bands is determined by the band gap width (E_g). To facilitate conductivity, such a material is doped. To achieve n-type conductivity (electron conductivity, negative charges), it is doped with elements from group V of the periodic table, such as phosphorus (P). P-type conductivity (hole conductivity, positive charges) is obtained by doping with elements from group III, such as boron (B). At the boundary of the n and p layers, a layer of space charge is formed - the p-n junction - which is crucial for the operation of the photovoltaic cell. The photovoltaic effect is a process in which light is converted into electrical energy. To be absorbed and generate a carrier pair, a photon must have an energy greater than or equal to the band gap width (E_g). For silicon, $E_g \approx 1.12$ eV. Absorption leads to the transition of an electron from the valence band to the conduction band. The electric field in the p-n junction acts as a separator. Immediately after generation, electron-hole pairs are separated. It is this process of generating and separating charges of different signs at opposite ends of the cell that creates the voltage. Total solar power at the average distance from Earth to the Sun, described as AM0 (Air Mass Zero) power, is approximately 1361 W/m^2 (the solar constant). However, after passing through the Earth's atmosphere, this power weakens, and at the Earth's surface it ranges from about 1200 W/m^2 at the tropics to about 900 W/m^2 in Central Europe, and about $500\text{-}700 \text{ W/m}^2$ at the poles (at the position of the Sun). These parameters are called AM1.5 (Air Mass 1.5) values, representing the spectral composition of light appropriately modified by the atmosphere. The irradiance (G) is a fundamental indicator of the solar power reaching the module. Short-circuit current (I_{sc}) is almost perfectly linearly proportional to irradiance. This relationship is described by the formula (1):

$$I_{sc} \approx \frac{I_{sc,STC} * G}{G_{STC}} \quad (1)$$

- the charge of an electron ($1.602 \cdot 10^{-19}$ C), V_{oc} – open circuit voltage, n - diode quality factor (for Si from 1 to 2), k - Boltzmann constant ($1.38 \cdot 10^{-23}$ J/K), T - cell temperature in Kelvin.

er (P) of the electric current delivered at a certain moment is the product of the and voltage (3), and the electric energy (E) is the integral of the current power e operation of the panel (4).

$$P(t) = I(t) \cdot V(t) \quad (3)$$

$$E = \int_{t=t_1}^{t_2} P(t) dt \quad (4)$$

stantaneous value of electric current, $V(t)$ - instantaneous value of electric voltage, t , t_1 , t_2 – time, start e of the system operation.

ous values of I and V are not exactly equal to those calculated from formulas (1) those given above are rather their upper estimates. The panels produce direct DC) with variable parameters. Connecting them to the power grid requires an at converts the DC to alternating current (AC) (see Fig. 1).

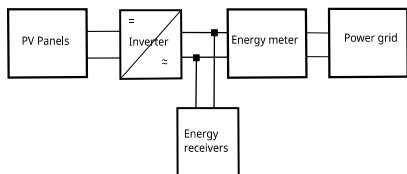


Fig. 1: Diagram of connecting PV panels and an inverter to the power grid.

y speaking, the values of t_1 and t_2 are determined by sunrise and sunset, but in y the switching on and off of the inverter. All the above formulas show the most t factors that influence the power of the panel, such as irradiance, length of day, the sun above the horizon, junction temperature, and the material from which it and more unpredictable factors such as cloud cover, dust, snow, hoarfrost, rain, ilar. It is not possible to examine their overall impact analytically, and ntal research is needed.

the hardware

ce consists of a measurement system controller and connected light (irradiation) temperature sensors. Access to the inverter is via a Wi-Fi router, which provides the local LAN and internet services (see Fig. 2). The main device of the system l-supported Raspberry PI minicomputer, which serves as the measurement controller. This device is quick and easy to configure thanks to prebuilt drivers. It both C and Python compilers. Both languages have the necessary libraries for cation via serial interfaces, such as I2C and 1-Wire. The device is connected to ia GPIO ports and their associated interfaces, as well as a built-in Wi-Fi module munication with a router and LAN. The minicomputer monitors the operation of ure and light sensors, collecting data at predefined intervals, and reads on results from the inverter. The inverter transforms the output voltage and also the operation of the photovoltaic system. Communication with the inverter is /IP using a local router, to which both the controller and the inverter are d. The local router also provides internet connectivity for remote services such , GitHub, and SSH.

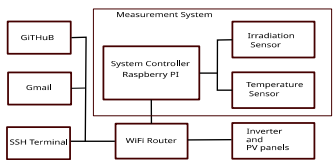


Fig. 2: System structure diagram

asurement system reads data from the sensors, collects a 24-hour data set, which oller saves to a CSV file kept locally on the SD card. The file is closed at , and a daily report is generated and sent to Gmail. Then, the measurement system e controller, logs into the LAN via Wi-Fi, and updates the software from GitHub. automatically launches the measurement script, and another daily cycle is

(22pcs.) G5 305-330

Tab. 1: Devices from which the system is built

the software

Measurement system software consists of a set of scripts run in a planned order, together with libraries on the Raspberry PI device. The project has been placed in GitHub repository. You can download its latest version from the [PVLogger.git](https://github.com/PVLogger) repository. This solution provides great flexibility in the development and creation of modifications. The measurement system has been programmed mainly in Python, C++ and shell commands to carry out individual subroutines in a certain logical order, enabling the implementation of the daily measurement cycle. Starting at midnight, the system executes scripts from MYSCRIPT.SERVICE containing scripts controlling the measurement system. The script AUTOSTART starts the internet connection, the software, and runs the main program. PVLOGGER is the main program that collects data in a loop at a specified time interval. Finally, each working day, the system sends the results and status notifications by email. Then, at midnight, it restarts the data again. The block diagram of the system operation is shown in Fig. 3.

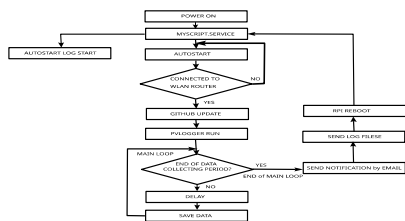
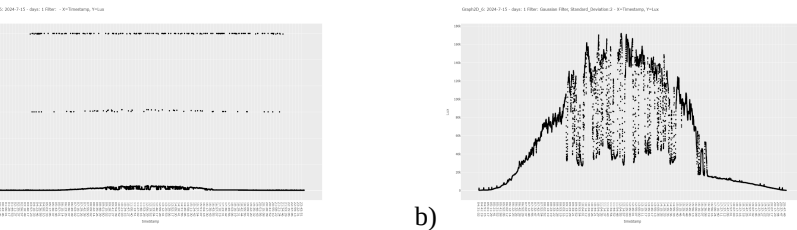


Fig. 3: The block diagram of the automatic measurement system operation

Information about the devices used can be found, in [Al26], [Df26], [Ra26], [Fr26], and [Mp26]. The sensor used measures illuminance in [lm], but the energy of the light is connected with irradiance. However, these quantities are linearly related for a given light source (solar light spectrum) and can be related using a formula: $685 \text{ Lux [lm/m}^2\text{]} = 1 \text{ W/m}^2$ or $1 \text{ Lux} = 0.0079 \text{ W/m}^2$. The maximum inverter power fed into the power grid. The maximum electrical power generated under STC conditions (Maximum Power Point).

lighting. Raw data collected from the sensor during one day (2024/07/15) are shown in Fig. 4a). Apart from the data series, one can notice two series that are erroneous because they contain data with impossible illuminance values. Therefore, the following data preprocessing was performed (results are shown in Fig. 4b). The erroneous data were deleted, and the remaining data were filtered to remove outliers using a Gaussian filter (5) (see [OS094]).



a) The noisy raw data obtained from the light sensor and b) the data after preprocessing using a Gaussian filter ($\sigma=2$)

$$G(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right) \quad (5)$$

value of the Gaussian function at point x , σ – standard deviation (controls the "blurring" of the filter), x – distance from the center (can be, for example, -1, 0, 1 for a 3-element filter).

The models

Regression model. The method of least squares (OLS – Ordinary Least Squares) consists of fitting a simple regression line/surface to minimize the sum of squares of differences between the observed and predicted values of the dependent variable. The coefficients of the regression surface (6) are determined from the formula (7).

$$Y = a_1 * x_1 + a_2 * x_2 + b \quad (6)$$

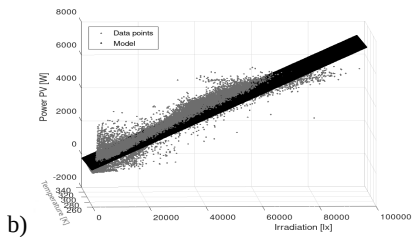
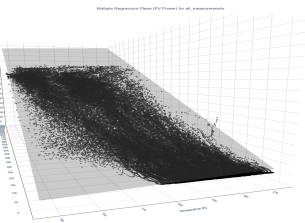
$$A = (X' * X)^{-1} * X' * Y \quad (7)$$

dependent data vector, and X is a matrix of independent data supplemented with a unit vector, $A = [a_1, a_2, b]$ – vector of searched regression parameters.

5.7580	0.077986	-1605.7	0.9424	2
3.5798	0.065990	-947.85	0.94643	78
1.1715	0.035135	-341.31	0.3700	13
4.9480	0.076804	-1401.0	0.9325	17
5.7610	0.083579	-1565.0	0.9731	24
4.1314	0.082172	-1084.8	0.9502	30
5.6454	0.083158	-1566.3	0.9622	85
-4.3980	0.075880	1348.1	0.9381	163

Summary of linear regression results for individual months, parts of 2024 and 2025, and all data (a_1 , a_2 , and b – regression parameters, R^2 – the coefficient of determination)

Model parameters presented in Tab. 2 are quite surprising, because while for all periods they are quite similar, especially the dependence on irradiance, for the test data set, the obtained model is different and, interestingly, more similar to the dependence of panel efficiency on temperature. Fig. 5 shows graphs presenting results of selected models.



Linear regression model obtained for all data, the dependence of electricity production a) on panel temperature; b) on solar radiation and temperature of panels

Neural network. The LSTM (Long Short-Term Memory) neural network (see [1], [CO17]). This enables deeper analysis of dependencies and introduces an element of memory or inertia, which is clearly present in the operation of PV panels. The structure of the LSTM network is described by formulas (8)–(11).

Fig. 6: Schematic diagram of the structure of one segment of the LSTM network

$$= \sigma_i(W_{x_i} \cdot x_t + W_{h_i} \cdot h_{t-1} + b_i) \quad i=1, 2, 3. \quad (8)$$

$$= \tanh_1(W_{x_4} \cdot x_t + W_{h_4} \cdot h_{t-1} + b_4) \quad (9)$$

$$= s_1 \cdot c_{t-1} + s_2 \cdot s_3 \quad (10)$$

$$= y_t = s_4 \cdot \tanh_2(W_{c_t} \cdot c_t + b_{c_t}) \quad (11)$$

weights for the input signal; W_{h_i} , W_{h_4} , W_{c_t} – weights for the short-term and long-term signal; b_i , b_4 , hold values (biases); s_i , s_4 – auxiliary symbols for the output signals from the corresponding function modules; c_t – long-term signal; h_t – short-term input signal; t , $t-1$ – subsequent time

t values were: irradiance, panel temperature, time, day of the week, and day of The vector of standard output signals was the PV power production. The data was appropriately preprocessed normalized, and divided into training and validation parts. The data selected for and testing the considered models were selected randomly, but using data from e days. The key challenge is selecting appropriate network parameters and procedures. The well-chosen parameters yield good results with an R^2 value an 0.99. Results obtained using the LSTM network are shown in Fig. 7. The now that the model obtained using the LSTM network is very accurate.

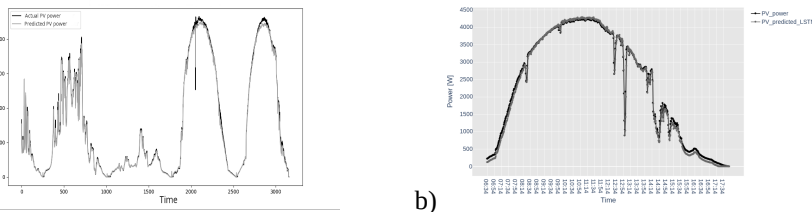


Figure 7: Comparison of real and obtained results, a) real data and test data; b) LSTM predictions for March 21, 2025.

ating the enormous potential of using it for modeling.

$$S_p = \frac{(E_r - |E_p - E_r|) * 100 \% }{E_r}$$

(12)

prediction accuracy, E_r – produced energy, E_p - prediction of the evaluated model.

	Energy [kWh]	Prediction: OLS [kWh]	Prediction: LSTM [kWh]	Accuracy (S) OLS [%]	Accuracy (S) LSTM [%]
24	32.195	34.792	31.302	91.934	97.226
24	5.094	5.599	4.68	90.086	91.873
25	4.028	5.375	3.689	66.559	91.584
25	26.621	23.231	26.026	87.266	97.765

Tab. 3: Prediction results of daily electricity production for the training data.

	Energy [kWh]	Prediction: OLS [kWh]	Prediction: LSTM [kWh]	Accuracy (S) OLS [%]	Accuracy (S) LSTM [%]
25	40.911	32.546	39.769	79.355	97.209
25	30.513	24.701	28.717	80.952	94.113
25	29.960	24.142	28.406	80.581	94.813

Tab. 4: Prediction results of daily electricity production for the testing data.

mmmary

developed device allows for estimating production data from any time range and can act as a predictor to estimate energy loss in the event of a forced shutdown. It is also possible to use the model as a predictor of energy production for a specific time period in the future. In this case, the forecast of irradiance and temperature for that time period is required. These data can be supplied either from external sources or estimated using a weather prediction model. It should be emphasized, that the system is quite simple, requiring only two basic sensors.

- Charu, A.: *Neural Networks and Deep Learning*, Springer, Hamburg, Germany, 2023.
- Cohen, J., et al.: *Applied multiple regression/correlation analysis for the behavioral sciences*, Hillsdale, NJ: Lawrence Erlbaum Associates, 2003.
- Dragomir, F., Dragomir, O.: *Photovoltaic Power Forecasting with AI: A Cost–Benefit Framework Across Multiple Time Horizons*, *Future Internet* 18(6):291, MDPI, 2026.
- Dyndał, K., Lewińska, G., Marszałek K.: *Fotowoltaika*, Kraków Open AGH e-Podręczniki, 2024, <https://epodreczniki.open.agh.edu.pl/handbook/35>, accessed 2026/04/13
- Gers, F., Schmidhuber, J., Cummins, F.: *Learning to Forget: Continual Prediction with LSTM*, *Neural Computation*, 12 (10): 2451–2471, (2000)
- Alldatasheet, www.alldatasheet.com/datasheet-pdf/pdf/58557/DALLAS/DS18B20.html, accessed 2026/04/14
- Dfimg, dfimg.dfrobot.com/wiki/18648/SEN0390_b-lux-v30b-ambient-light-sensor_datasheet_V1.0.pdf, accessed: 2026/04/14
- Raspberry Pi, www.raspberrypi.com/documentation/computers/os.html#gpio-and-the-40-pin-header, accessed 2026/04/14
- Fronius, www.fronius.com/en-gb/uk/search-page?searchword=fronius-symo-3-0-3-m&activeDepartment=solarenergy&categoryFacet=category-downloads, accessed 2026/04/14
- Mpe-online, mpe-online.ie/wp-content/uploads/2018/11/Hanwha_Q_CELLS_Data_sheet_QPEAK_DUO-G5_305-330_2017-10_Rev01_EN.pdf, accessed 2026/04/14
- Oppenheim, A., Schafer, R.: *Discrete-Time Signal Processing*, Pearson, 2009.